

MAUN, Botswana

WMO: 680320

Lat: 19.973S

Lon: 23.431E

Elev: 943

StdP: 90.50

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	8.5	-9.0	2.0	18.2	-6.4	2.5	18.7	8.3	18.6	7.4	19.2	2.3	140	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	14.8	38.1	18.8	37.1	18.8	36.1	18.9	23.6	29.8	23.2	29.5	22.8	29.0	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.9	25.1	21.9	18.6	24.9	21.2	17.8	24.6	76.0	29.9	74.2	29.7	72.7	29.1	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.4	DB	4.1	40.3	1.7	1.5	2.8	41.4	1.8	42.3	0.9	43.2	-0.4	44.3	
			WB	0.1	25.1	1.6	0.8	-1.0	25.7	-1.9	26.2	-2.8	26.7	-4.0	27.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	26.7	26.6	25.7	24.0	20.7	18.1	17.6	20.8	25.2	28.2	27.6	27.0
	DBStd	4.24	2.12	1.64	2.12	2.00	2.35	2.12	2.02	2.72	2.58	2.36	2.73	2.11
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	83	0	0	0	1	6	29	39	8	0	0	0	0
	CDD10.0	5108	517	466	486	420	333	243	236	336	456	564	527	526
	CDD18.3	2148	258	232	227	170	81	22	16	86	206	306	277	268
	CDH23.3	24753	2517	2254	2173	1747	1049	469	441	1209	2637	4126	3391	2739
Wind	WSAvg	3.2	3.0	2.9	3.0	3.0	3.0	2.9	3.1	3.3	3.6	3.8	3.4	3.1
	PrecAvg	451	109	97	58	26	4	1	0	0	5	16	48	79
	PrecMax	925	347	366	193	120	34	17	5	10	32	101	112	262
	PrecMin	222	20	7	0	0	0	0	0	0	0	0	2	13
	PrecStd	161	70	84	50	30	8	3	1	2	8	23	34	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.9	37.0	37.1	34.9	33.1	30.6	30.2	34.0	37.4	39.2	39.8	37.8
		MCWB	19.6	20.0	19.8	19.6	17.6	15.7	14.3	16.5	17.5	18.8	19.5	19.8
	2%	DB	35.9	35.3	35.2	33.6	31.4	29.1	28.8	32.4	36.1	38.1	37.9	36.0
		MCWB	21.2	20.9	19.9	18.9	17.1	15.6	14.2	15.8	17.3	18.2	19.5	20.3
	5%	DB	34.1	34.1	33.9	32.2	30.1	27.8	27.2	31.1	35.1	37.2	36.2	34.7
		MCWB	21.4	20.9	20.1	18.8	16.5	14.8	13.8	15.3	16.9	18.1	19.3	20.8
	10%	DB	32.8	32.9	32.0	31.0	28.8	26.2	26.0	29.8	34.0	36.1	35.0	33.1
		MCWB	21.5	21.2	20.3	18.9	16.3	14.4	13.5	14.8	16.5	18.0	19.5	21.1
	0.4%	WB	24.4	24.1	23.7	22.7	20.5	18.0	17.0	19.1	20.7	22.1	23.0	24.0
		MCDB	30.9	29.7	28.7	29.7	28.8	27.6	27.0	30.5	33.3	34.6	31.9	30.2
	2%	WB	23.6	23.4	23.1	21.8	19.1	16.9	15.5	17.3	19.3	21.1	22.1	23.2
		MCDB	30.0	29.6	28.5	28.4	27.1	26.4	26.5	29.4	32.4	33.1	30.9	29.8
	5%	WB	23.2	23.0	22.6	21.0	18.0	15.8	14.6	16.2	18.3	20.3	21.6	22.7
		MCDB	29.4	29.4	28.0	27.4	27.0	25.4	25.6	29.0	31.2	31.9	30.2	29.3
	10%	WB	22.7	22.6	22.1	20.3	17.0	14.9	13.8	15.3	17.5	19.6	21.1	22.2
		MCDB	28.5	28.6	27.5	27.0	26.6	24.7	24.7	28.1	30.7	31.2	29.6	28.8
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	11.8	14.1	16.0	16.5	16.9	17.0	16.7	14.8	13.2	11.4
		MCDBR	13.6	14.1	14.7	16.0	17.0	18.0	18.7	18.0	17.7	16.2	15.3	13.9
		MCWBR	4.3	4.4	5.2	6.3	7.5	8.5	9.0	8.0	7.5	6.3	5.4	4.2
	5% WB	MCDBR	10.4	10.8	10.7	12.9	15.3	15.9	16.9	16.9	16.5	14.3	12.6	11.0
		MCWBR	3.7	3.7	4.0	5.1	7.1	8.0	8.7	8.1	7.6	5.6	4.4	3.8
Clear-Sky Solar Irradiance	taub		0.383	0.383	0.367	0.336	0.303	0.298	0.303	0.371	0.454	0.447	0.398	0.382
	taud		2.400	2.407	2.437	2.464	2.500	2.474	2.440	2.243	2.032	2.107	2.303	2.393
	Ebn at Noon		961	949	940	931	928	913	918	881	844	880	941	963
	Edh at Noon		128	125	117	105	94	92	98	129	172	167	140	129
All-Sky Solar Radiation	RadAvg		6.75	6.44	6.06	5.67	5.21	4.69	4.92	5.64	6.35	6.86	6.74	6.65
	RadStd		0.53	0.55	0.44	0.37	0.15	0.19	0.14	0.22	0.30	0.41	0.56	0.39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page